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Date:

**MAI 02, 2025**

# TECHNICAL REPORT

**Sentinel-1 Mining Area Change Detection Over Al Fujairah.**


## Confidentiality Notice:

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Prepared For:

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هيئة الفجيرة للبيئة
FUJAIRAH ENVIRONMENT AUTHORITY

# Summary

This report presents the results of satellite-based change detection analysis focused on the Al Fujairah region. The objective was to identify mining area expansion across multiple years using time-series satellite imagery.

The analysis covered the following time intervals:

- 2016 → 2019
- 2019 → 2021
- 2021 → 2023
- 2023 → 2025
- 2016 → 2025

The total area under observation is approximately 52 km². Based on visual comparison and interpretation of multi-year imagery, significant landscape changes were identified, consistent with expanded mining activity. These include deforestation, terrain leveling, and new excavation zones

# Data Specifications:

**Data Specifications**
**Location:** 25.557586° N, 56.004974° E
**Area Covered:** 52 km²
**Analysis Period:** 2016 to 2025 (in 2–3-year intervals).

**Satellite Sources:**

- Open-access Earth Observation Data
- High-resolution optical imagery

**Methodology:**

- Multi-temporal image acquisition
- Visual and pixel-based change detection
- Validation using geolocation coordinates

**Figure Description (Applies to Figures 1–5)**
Each figure consists of four panels labeled **(a) to (d)** to illustrate the progression and detection of mining changes over time:

- **(a) Satellite Image – Year 1 (e.g., 2016)**
  Displays the original landscape prior to mining expansion.

- **(b) Satellite Image – Year 2 (e.g., 2019)**
  Shows the same region after a few years, capturing visual differences.

- **(c) Change Detection Map (Masked Overlay)**
  Highlights detected mining activity between the two years.
  Colored areas represent new or expanded mining zones, with the total changed area measured in square kilometers (km²).

- **(d) Top Change Clusters (Colored Segments)**
  Identifies and separates the largest change areas using different colors.
  Each segment is labeled with the estimated area affected (in km²), helping to quantify and compare mining intensity across the site.

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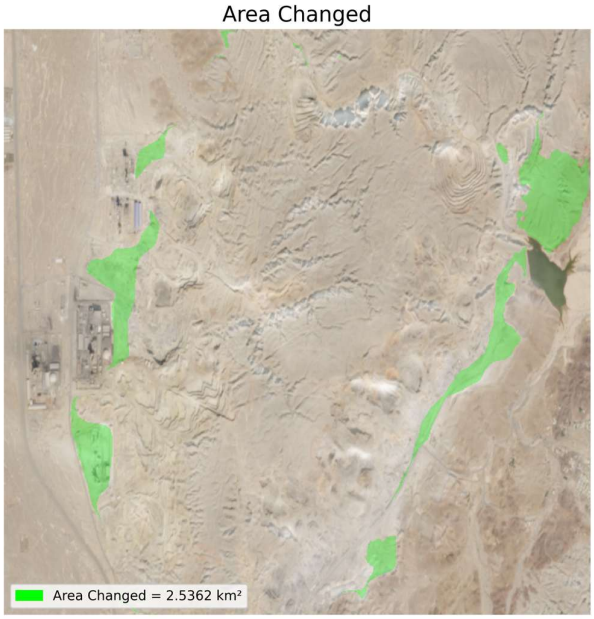
**Visual Analysis**

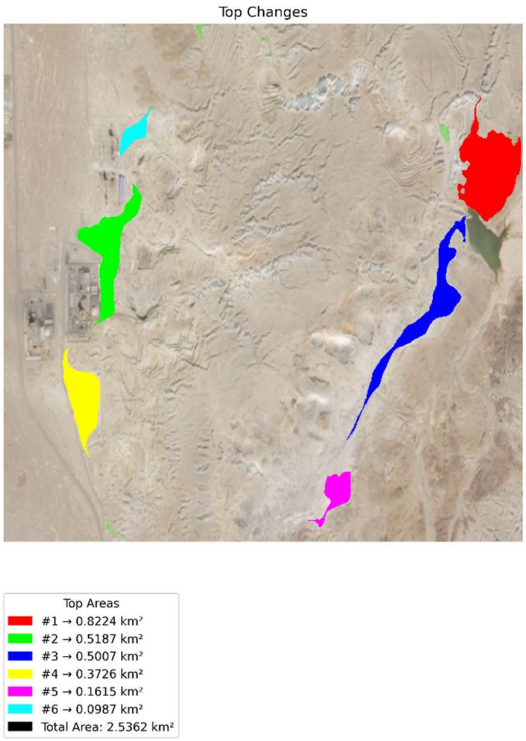
**Figure 1: Visual Comparison (2016 → 2019)**

(a) Satellite Image – Year 2016

(b) Satellite Image – Year 2019



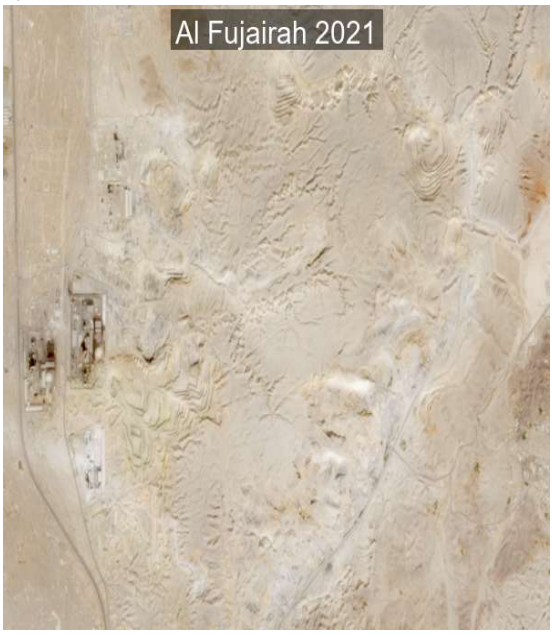

(c) Change Detection Map (Masked Overlay)




(d) Top Change Clusters (Colored Segments)

**Figure 2: Visual Comparison (2019 → 2021)**

(a) Satellite Image – Year 2019




(b) Satellite Image – Year 2021

(c) Change Detection Map (Masked Overlay)

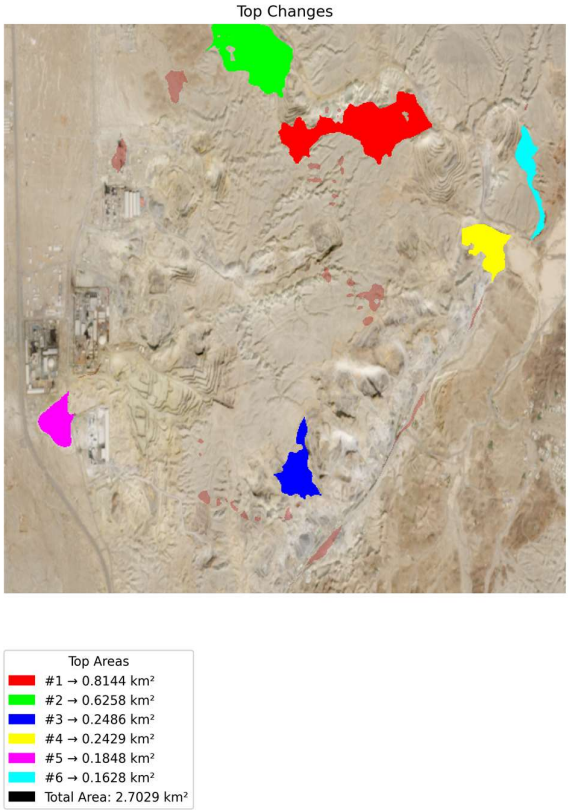
(d) Top Change Clusters (Colored Segments)

**Figure 3: Visual Comparison (2021 → 2023)**

(a) Satellite Image – Year 2021

(b) Satellite Image – Year 2023

(c) Change Detection Map (Masked Overlay)



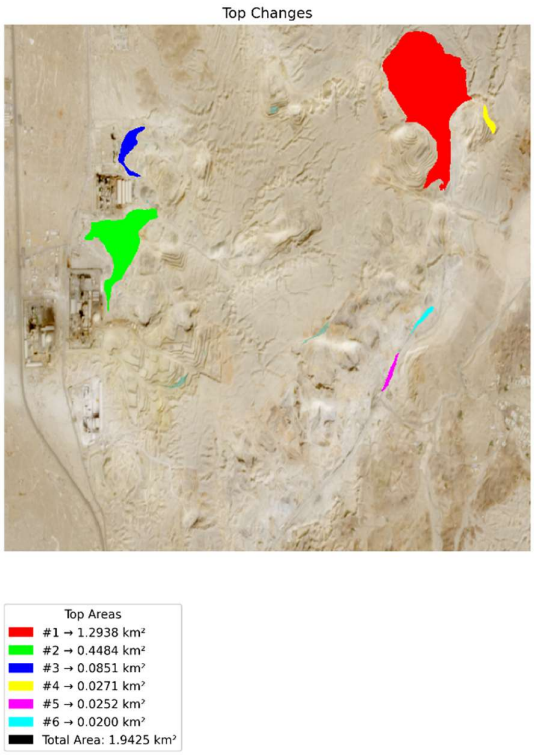

(d) Top Change Clusters (Colored Segments)

**Figure 4: Visual Comparison (2023 → 2025)**

(a) Satellite Image – Year 2023

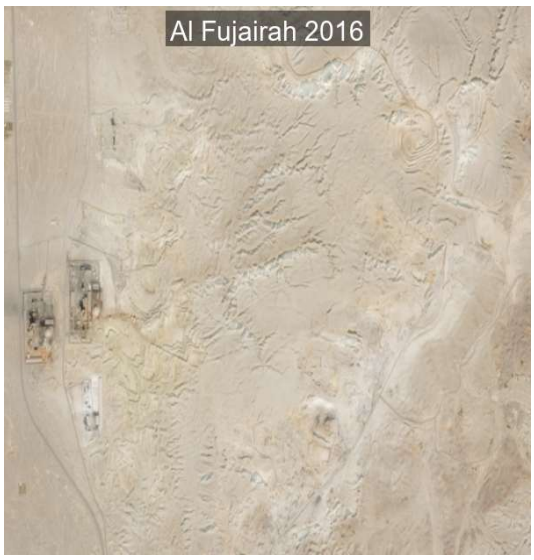
(b) Satellite Image – Year 2025

(c) Change Detection Map (Masked Overlay




(d) Top Change Clusters (Colored Segments)

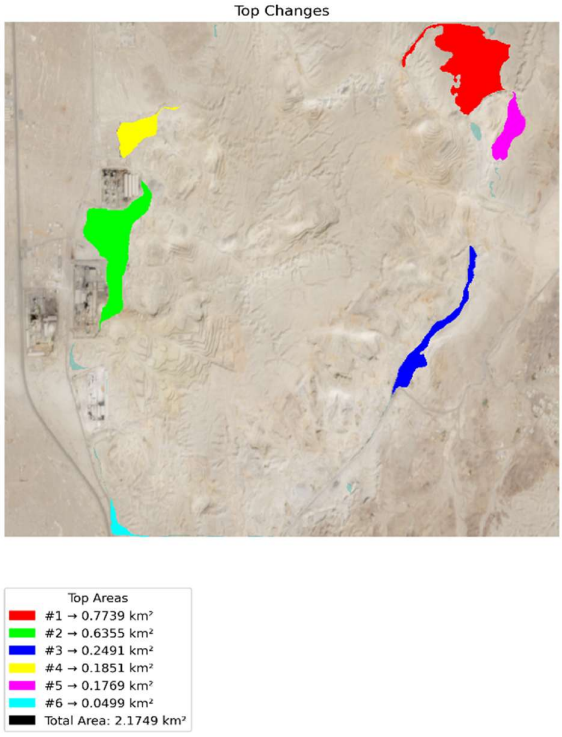
**Figure 4: Visual Comparison (2016 → 2025)**




(a) Satellite Image – Year 2016

(b) Satellite Image – Year 2025

(c) Change Detection Map (Masked Overlay)




(d) Top Change Clusters (Colored Segments)

*Note: These images illustrate progressive changes in land cover and terrain structure, indicating intensified mining activity.*

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**Conclusion**

The analysis confirms a consistent pattern of mining expansion in the Al Fujairah area from 2016 to 2025. Such changes are critical to monitor for environmental management, compliance, and land use planning.

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# Thank You

We look forward to potential collaboration.


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